

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.1: Similarity and Enlargement — Teacher Reference (Pre-filled)

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Driving Question	DRIVING QUESTION: Why does doubling the size of a container make it hold eight times as much — and how can we use this mathematical pattern to design, build, and predict the properties of any similar shape or object?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Noticing Similarity — What Makes Two Shapes the "Same but Different"?	Three uji cups of different sizes (small, medium, large) share the same shape — every dimension of the medium cup is exactly twice that of the small cup ($k = 2$) and every dimension of the large cup is exactly three times that of the small cup ($k = 3$). When filled with water, the medium cup required 4 small cups (not 2) and the large cup required 9 small cups (not 3). In everyday objects measured by groups, similar pairs showed equal corresponding angles and a constant ratio of corresponding sides.	Two figures are similar if and only if their corresponding angles are equal AND their corresponding sides are in a constant ratio called the linear scale factor (k). Similarity is about shape, not size. The linear scale factor k can be found by dividing the length of any side of the larger figure by the corresponding side of the smaller figure. For the uji cups, $k = 2$ (medium:small) and $k = 3$ (large:small) — but the number of water fills was $k^2 = 4$ and $k^2 = 9$, not k itself. This k^2 pattern is the mystery that the rest of the unit will explain.	The three uji cups are similar solids — they share the same shape, with all dimensions scaled by the same factor k . This is why they look proportional to the eye. However, the amount of water each holds (related to cross-sectional area and volume) does not scale by k but by a larger power of k . Today's lesson establishes the foundation — the definition of similarity and the meaning of k — which is the first step toward building a mathematical model that predicts the 'surprising' water-fill numbers and explains why scaling lengths by k scales areas and volumes by higher powers of k .
Lesson 2 Centre of Enlargement and Linear Scale Factor	Learners traced a pentagon ABCDE and its enlarged image on plain paper, drew rays through three pairs of corresponding vertices ($A \rightarrow A'$, $B \rightarrow B'$, $C \rightarrow C'$), and identified the single point O where all rays intersected as the centre of enlargement. They also measured five pairs of corresponding sides on the pentagon and computed the ratio image÷object for each pair.	The centre of enlargement O is the unique fixed point from which every vertex of the object is projected outward to its corresponding image vertex. The linear scale factor $k = \text{image length} \div \text{object length}$ is constant for ALL pairs of corresponding lengths in an enlargement — it does not vary from side to side. For the uji cups, $k = 2$ (small $\rightarrow$ medium) and $k = 3$ (small $\rightarrow$ large), meaning every single length	Because k is the ratio of ANY matching pair of lengths, the manufacturer can guarantee that the large uji cup is geometrically similar to the small one simply by using the same centre of enlargement and the same scale factor k for every point. This is why the cups 'look the same' despite being different sizes. However, k only describes lengths — the class has not yet explained why filling the cup requires k^2 times as much water in

		on the medium cup is exactly twice the matching length on the small cup.	cross-section, or k^3 times as much in volume. Those ??? boxes in the Anchor Model will be resolved in Lesson 3.
Lesson 3 Drawing Enlargements — Constructing Images from a Centre and Scale Factor	Learners constructed enlargements of triangles and rectangles using three different scale factors ($k = 2$, $k = 3$, $k = \frac{1}{2}$) from a centre of enlargement O, then measured every pair of corresponding sides and angles to verify their constructions. They saw that the image is always farther from O (for $k > 1$) or closer to O (for $k < 1$), and that the verification table consistently showed the length ratio equal to k and the angles equal.	Under enlargement by scale factor k : (1) every length in the image equals k times the corresponding object length; (2) every angle in the image equals the corresponding object angle — shape is preserved; (3) the centre of enlargement O remains fixed. A scale factor $k > 1$ enlarges; $0 < k < 1$ reduces. These results hold for ANY pair of corresponding lengths — not just selected sides.	This explains part of the uji cup phenomenon: because the cups are enlargements of each other, every single length on the medium cup is exactly $k = 2$ times the matching length on the small cup. The circular cross-section has both its radius AND its height scaled by $k = 2$. Since area involves two length dimensions multiplied together, the area scales by $k \times k = k^2 = 4$ — which is exactly why four small cups of water were needed to fill the medium cup's cross-section, not two.
Lesson 4 Area Scale Factor: Why Does the Uji Cup Label Need Nine Times as Much Paper?	Learners measured and compared the areas of sets of similar squares and rectangles (with $k = 2, 3, 4$) cut from manila paper, recorded their data in a structured table, and plotted area ratio against k on a graph.	The area scale factor of similar figures equals k^2 (the square of the linear scale factor), because both length and width are each multiplied by k , making the total area grow by $k \times k$. This explains why the cross-sectional area of the medium uji cup ($k = 2$) is 4 times — not 2 times — the cross-section of the small cup.	Using the k^2 rule, learners explained and predicted how the cross-sectional area of a similar shape changes when the linear scale factor is known, and applied this to real-world contexts including shamba areas in Kericho, billboard paint coverage in Nairobi, and triangular road signs.
Lesson 5 Volume Scale Factor: Why the Large Uji Cup Holds Eight Times as Much	Learners measured the dimensions of similar solids (model cubes/cuboids or Kasuku tins with $k = 2$ and $k = 3$) and computed their volumes. They pooled class data in a shared table showing the measured volume ratio alongside k^2 , and k^3 for each scale factor.	The volume scale factor of similar solids equals k^3 , where k is the linear scale factor. This means that when every length is scaled by $k = 3$, the volume becomes $3^3 = 27$ times as large. The complete chain is now established: length scales by k , area by k^2 , and volume by k^3 .	Learners explained the anchoring phenomenon fully: the large uji cup ($k = 3$) needs 27 small cups of water — not 3 and not 9 — because volume is a three-dimensional quantity and each of the three dimensions is multiplied by k , giving $k \times k \times k = k^3$. They also applied this to the sufuria supporting phenomenon, confirming that a sufuria with $k = 2$ holds $2^3 = 8$ times as much ugali porridge as the small sufuria.
Lesson 6 Relating Linear, Area and Volume Scale Factors — The Unified k, k^2, k^3 Model	Two similar cylindrical water tanks from the Rift Valley with linear scale factor $k = 3$; a unified $k / k^2 / k^3$ table completed and verified using digital devices; the GeoGebra/Kenya Education Cloud simulation showing surface area and volume changing as cylinder dimensions are scaled by $k = 2, 3, 4$, and 5.	For any two similar figures or solids with linear scale factor k : corresponding lengths scale by k , corresponding surface areas scale by k^2 , and corresponding volumes scale by k^3 . For $k = 3$: the large Rift Valley tank needs 9 times more paint ($k^2 = 9$) and holds 27 times more water ($k^3 = 27$). This single rule — one value of k encoding three scaling behaviours — explains every surprising number observed in the Shrinking Uji Cup phenomenon since	The anchoring phenomenon is now fully explained: when the uji cup manufacturer doubled every dimension ($k = 2$), the surface area of the label increased by $k^2 = 4$ (four times more paper) and the volume increased by $k^3 = 8$ (eight small cups needed to fill it, not two). When dimensions were tripled ($k = 3$), the label needed $k^2 = 9$ times more paper and the volume needed $k^3 = 27$ times more water (nine small cups, not three). The driving question — 'Why

		Lesson 1.	does doubling the size of a container make it hold eight times as much?' — is answered: because volume scales as the cube of the linear scale factor, so $k = 2$ gives $k^3 = 8$.
Lesson 7 Real-Life Applications and Problem Solving — Apply & Communicate	Learners measured scale drawings of a school building, calculated flour quantities for a scaled-up mandazi tin ($k = 2$, requiring $8\times$ the dough), and used a Kisumu–Kericho road map at scale 1:500 000 to calculate real distances and compare map areas at a different scale.	The linear scale factor k governs lengths, k^2 governs areas (including surface area and map area), and k^3 governs volumes (including recipe quantities for similar containers). These three rules apply consistently across all real-life domains — architecture, catering, cartography — involving similar figures or solids.	The uji cup phenomenon — where doubling every dimension of the cup requires $8\times$ as much water to fill it, not $2\times$ — is explained by the k^3 volume rule. Because volume depends on three linear dimensions multiplied together, scaling each by k scales the total volume by $k \times k \times k = k^3$. The same logic explains why Mama Akinyi needed 4 000 g of flour (not 1 000 g) for the large mandazi tin, and why the Lake Victoria area on a redrawn map at twice the linear scale is $4\times$ larger, not $2\times$.
Lesson 8 Project Presentations, Final Explanation & DQB Completion	Learners observed their own peer-constructed similar containers (cubes, cuboids, cylinders) made from locally available materials in at least two sizes. They measured corresponding dimensions side by side, saw with their own hands that every length scaled by the same factor k , and verified (or discovered errors in) their calculated surface area ratios and volume ratios. They also re-observed the three uji cups from Lesson 1 — now able to explain precisely why nine small cups of water, not three, are needed to fill the large cup.	Learners consolidated that: (1) similar figures have equal corresponding angles and proportional corresponding sides; (2) the linear scale factor k is the ratio of any pair of corresponding lengths; (3) the area scale factor is always k^2 — because area involves two linear dimensions each scaled by k ; (4) the volume scale factor is always k^3 — because volume involves three linear dimensions each scaled by k ; (5) these relationships apply universally to all similar figures and solids, including the containers they designed and built themselves; (6) real-life applications include container manufacturing, architectural scale models, map-making, and packaging design.	Learners explained — through project presentations, peer rating, collaborative DQB closure, final model annotation, and individual written tasks — why doubling all linear dimensions of any container multiplies its surface area by 4 ($= 2^2$) and its volume by 8 ($= 2^3$). They connected this explanation back to the anchoring phenomenon: the large uji cup has $k = 3$ relative to the small cup, so its cross-sectional area is $k^2 = 9$ times larger (explaining why nine small cups fill it) and its volume is $k^3 = 27$ times larger. They demonstrated this understanding using their own constructed containers and novel Kenyan contexts (Rift Valley water tanks, Nairobi label design, mandazi moulds), satisfying the KICD key inquiry question: How is similarity and enlargement applied in real life?

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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